Signed multichannel reconstruction on a factorial analytic cylinder domain

Project Stardust

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Abstract

For the signed heat kernel of even order $p = 2m$, we construct a multichannel reconstruction formula on a specified Banach completion of dyadic polynomial cylinders. The coefficient fields are genuine Bochner L^1 limits of native predictable Taylor coefficients and have degree-independent bounds on the original observable norm. The centered observable pairing is the sum of $p-1$ bounded signed channels in the continuous test dual. The proof begins with an exact finite signed telescope, proves quantitative increment and adjacent-refinement estimates, and only then passes to the observable completion. A joint diagonal realizes every completed observable through actual signed polynomial sums. The pairing is indefinite; no positive path-space measure or stochastic isometry is asserted. We distinguish canonical coefficient uniqueness from uniqueness of channel synthesis. All principal results are supported by a frozen Lean development. Novelty relative to earlier higher-order signed calculus remains unresolved.

1 Scope and normalization

Fix $m \in \mathbb{N}$, $m > 0$, a physical horizon $T > 0$, and put

$$p = 2m, \qquad c = (-1)^{m+1}.$$

The fundamental signed kernel k_τ, $\tau > 0$, has Fourier transform $e^{-\tau\xi^p}$ and generator $c\partial_x^p$. We normalize the Fourier transform by

$$k_\tau(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\tau\xi^p} e^{-i\xi x}\, d\xi.$$

Changing the Fourier sign gives the same even real kernel. For each finite partition, independent increments mean the product of these signed densities; they do not mean independent random variables under an assumed positive path law.

The analytic domain $\mathcal{D}$ will be defined explicitly in Section 3. It has a dense injective polynomial core, a bounded signed mean $\mathcal{E}_T : \mathcal{D} \to \mathbb{R}$, and a bounded symmetric pairing

$$J_T : \mathcal{D} \longrightarrow \mathcal{D}^*, \qquad J_T(P)(G) = \mathcal{E}_T(PG)$$

on polynomial cylinders. Products in this last identity are finite polynomial products; bounded multiplication on the whole completion is not being assumed. Let λ denote Lebesgue measure on $[0,1]$, so time in the coefficient fields is normalized time and the physical horizon occurs in all kernel parameters and channel constants. Write $\mathcal{F} = L^1(\lambda; \mathcal{D}^*)$ for Bochner L^1.

Theorem 1.1. *There are canonical bounded linear maps $C_q : \mathcal{D} \to \mathcal{F}$, $q \geq 0$, satisfying*

$$\|C_q u\|_{\mathcal{F}} \leq \frac{\|J_T\|\, e^T}{q!}\, \|u\|_{\mathcal{D}}\,. \tag{1}$$

For $1 \leq q \leq p-1$, define

$$(I_q A)(v) = cpT(p-1)_{q-1} \int_0^1 A(s)(D_s^{p-q} v)\, ds, \tag{2}$$

where $(r)_k = r(r-1)\cdots(r-k+1)$ and D_s is the bounded cutoff derivative defined below. Then every $u \in \mathcal{D}$ satisfies

$$J_T(u - \mathcal{E}_T(u)1) = \sum_{q=1}^{p-1} I_q(C_q u) \qquad \text{in } \mathcal{D}^*. \tag{3}$$

On every fixed polynomial cylinder, C_q is the norm limit in $\mathcal{F}$ of its native predictable Taylor coefficient fields. Its values lie in the closed span of actual adapted polynomial step fields. Every completed observable admits a simultaneous polynomial/mesh approximation of all surviving fields and the right side of (3).

The equality is in the test dual, not an assertion of equality in $\mathcal{D}$ obtained by inverting J_T. Injectivity of J_T on the entire completion is not supplied. The theorem is therefore a signed Clark–Ocone-type reconstruction with its topology specified.

Higher-order signed heat calculus has substantial antecedents, including Hochberg's work [1]. Complex random-walk constructions give another generalized calculus [2], while time-changed pseudoprocesses are studied in [3]. The present paper does not claim the first higher-order Itô calculus. Its precise subject is the simultaneous native coefficient limit and bounded observable completion for these actual signed cylinder laws. Priority for this package remains unresolved.

2 Actual finite signed laws and polynomial conditioning

The function $e^{-\tau\xi^p}$ is Schwartz. Fourier inversion shows that k_τ is real, even and Schwartz, has integral one, and satisfies $k_\tau * k_\sigma = k_{\tau+\sigma}$. Thus polynomial functions are integrable against its total variation. Its moments are

$$\mu_r(\tau) = \int x^r k_\tau(x)\, dx = \begin{cases} (pk)!(c\tau)^k/k!, & r = pk, \\ 0, & p \nmid r. \end{cases} \tag{4}$$

For $r = 0$ this gives one. To obtain the formula, differentiate the Fourier transform at zero; Schwartz decay justifies differentiation under the integral. The derivative of $e^{-\tau\xi^p}$ vanishes unless the order is a multiple of p, and the resulting powers of i give c^k. Equivalently, for every polynomial f,

$$\int f(x) k_\tau(x)\, dx = (e^{c\tau\partial^p} f)(0). \tag{5}$$

The exponential series terminates on each polynomial.

An integration-by-parts identity that will control all power channels is

$$\int x f(x) k_\tau(x)\, dx = cp\tau \int f^{(p-1)}(x) k_\tau(x)\, dx. \tag{6}$$

It follows either from the moments (4), by testing monomials, or from differentiation of the Fourier transform. All boundary terms vanish by Schwartz decay.

On the dyadic grid with $N = 2^n$ cells set $h = 1/N$, $\tau = Th$, and use increment variables $x_0, \dots, x_{N-1}$. The signed expectation $\mathcal{E}_{T,n}$ integrates against $\prod_j k_\tau(x_j)\, dx_j$. The position variables are $y_i = x_0 + \cdots + x_i$. Refinement replaces each coarse increment by the sum of two fine increments. Convolution implies consistency of these actual signed expectations.

For $0 \le j \le N$, let $\Pi_j P$ integrate increments with indices $i \ge j$, leaving $x_0, \dots, x_{j-1}$ untouched. It is a polynomial in the retained increments. In particular $\Pi_0 P = \mathcal{E}_{T,n} P$, $\Pi_N P = P$, and $\Pi_j \Pi_k = \Pi_{\min(j,k)}$. These formulas follow from finite signed Fubini, with absolute integrability of polynomials against the product variation. The notation is polynomial conditioning; no general conditional expectation theorem for an indefinite path law is invoked.

3 The common factorial domain

Let S be the union of the positive dyadic grid times in $(0, 1]$. A cylinder is a polynomial in finitely many formal position coordinates Y_s, $s \in S$. The direct-limit construction using increment refinement gives exactly this polynomial algebra: old positions are retained at their original times, and the cumulative coordinate change is invertible on each finite grid.

For a finite multiindex $\alpha : S \to \mathbb{N}$, let $|\alpha| = \sum_s \alpha_s$. For a polynomial $P = \sum_\alpha a_\alpha Y^\alpha$, define

$$\|P\|_{\mathcal{D}} = \sum_\alpha |\alpha|!\, |a_\alpha|. \tag{7}$$

This is the factorial of total degree, not the product of coordinate factorials. The coefficients here are in *position* coordinates, after converting a given increment polynomial by the inverse cumulative substitution. Refinement is an isometry because it only retains the same position labels. It also does not increase total degree. The norm is nondegenerate because polynomial coefficients are unique.

Define $\mathcal{D}$ as the completion of this normed polynomial space. More concretely, it is the closed span of the vectors $(|\alpha|! a_\alpha)_\alpha$ in ℓ^1 indexed by finitely supported time multiindices. The Lean construction uses all times in $(0, 1]$ as an ambient index and takes the closed span of the dyadic core. This yields the same space just described. It is a specified analytic coefficient completion, not a space of all bounded smooth path observables.

Lemma 3.1. *For any direction $v : S \to \mathbb{R}$ with $|v_s| \le 1$, the polynomial directional derivative $\partial_v = \sum_s v_s \partial_{Y_s}$ extends to $\mathcal{D}$ with norm at most one. Position substitutions that send each variable to another variable or zero, without scaling by a coefficient of modulus greater than one, are contractions. The finite signed projection Π_j has norm at most e^T on each cylinder norm.*

Proof. For a monomial of degree $d > 0$, the ratio of the weighted norm of its directional derivative to $d!$ is at most $\sum_s \alpha_s / d = 1$. The degree-zero derivative vanishes. Summation proves the derivative bound. Substitutions preserve total degree or lower it; merging equal monomials can only reduce the sum of absolute coefficients.

In position coordinates, differentiation in the increment x_j is the direction which is one on positions after that increment and zero before it. Its norm is therefore at most one. Integrating one increment is heat evolution $e^{c\tau \partial_{x_j}^p}$ followed by setting that increment to zero. Setting it to zero merges later positions with their predecessors, so is contractive. The heat operator has norm at most e^τ by its exponential series. Composing future heat operators has norm at most $e^{\sum \tau} \le e^T$, and the remaining clamping substitution is contractive. □

Taking the empty-past projection gives a bounded signed mean $\mathcal{E}_T$ on $\mathcal{D}$, with $\|\mathcal{E}_T\| \le e^T$ and $\mathcal{E}_T 1 = 1$. The cutoff derivative in (2) is

$$D_s = \partial_{v(s)}, \qquad v(s)_t = 1_{\{s \le t\}}. \tag{8}$$

It is contractive, as are all its powers. Applied to a fixed polynomial, it is a finite step function of s. Approximation by core polynomials shows $s \mapsto D_s^r u$ is strongly measurable for every $u \in \mathcal{D}$: the applied fields converge pointwise, even uniformly in the norm bound. This gives a bounded map into $L^\infty(\lambda; \mathcal{D})$.

4 The product pairing and its signed character

Ordinary multiplication does not have norm at most the product of the norms in (7). Instead define on polynomials

$$B(P, Q)(Y) = P(Y/2)Q(Y/2).$$

For monomials of degrees d, e, its weighted norm ratio is $2^{-d-e}(d+e)!/(d!e!) \le 1$. Summing absolute coefficients gives $\|B(P,Q)\| \le \|P\| \, \|Q\|$, so B extends as a bounded symmetric bilinear map $\mathcal{D} \times \mathcal{D} \to \mathcal{D}$. Set

$$J_T(u)(v) = \mathcal{E}_{2^p T}\big(B(u, v)\big). \tag{9}$$

Then $\|J_T\| \le e^{2^p T}$. Homogeneity of the moments shows that half-dilation at horizon $2^p T$ restores horizon T: $\mu_r(2^p \tau) 2^{-r} = \mu_r(\tau)$. Applying this to each increment monomial proves

$$J_T(P)(G) = \mathcal{E}_T(PG) \tag{10}$$

for every pair of cylinders, and $J_T(u)(1) = \mathcal{E}_T u$ by continuity. This construction supplies the bounded product *pairing* without assuming bounded ordinary multiplication.

For $p = 4$ one has $c = -1$, so (4) gives

$$J_T(Y_1^2)(Y_1^2) = \mathcal{E}_T(Y_1^4) = -24T < 0.$$

Thus the pairing is indefinite. In particular the notation $\mathcal{E}_T$ cannot be used to justify positive Cauchy–Schwarz, martingale isometries, or Hilbert orthogonal projections.

For later estimates it is useful to name

$$W_j(H)(s) = 1_{(jh,(j+1)h]}(s) J_T(H).$$

This is an actual element of $\mathcal{F}$, with $\|W_j(H)\|_{\mathcal{F}} = h \, \|J_T(H)\|$. An adapted step field has this form summed over j, with H_j a polynomial depending only on increments before cell j. Let $\mathcal{A}_T$ be the closed linear span of all such fields in $\mathcal{F}$. This definition is the exact adaptedness space used below; it does not presuppose a probability filtration.

5 The bounded channel operators

For $A \in \mathcal{F}$ and $v \in \mathcal{D}$, the integrand $A(s)(D_s^r v)$ is integrable because its absolute value is at most $\|A(s)\| \, \|v\|$. Strong measurability follows from the bounded bilinear evaluation map and the applied-field measurability proved above. Define

$$Q_r(A)(v) = \int_0^1 A(s)(D_s^r v) \, ds.$$

Then $Q_r : \mathcal{F} \to \mathcal{D}^*$ is bounded with norm at most one. In addition to the surviving channels in (2), use the same formula for $q = p$. In that case

$$I_p = cp!TQ_0. \tag{11}$$

It is a drift channel, which will cancel exactly after centering. For $1 \le q \le p$,

$$\|I_q A\| \le pT(p-1)_{q-1} \|A\|_{\mathcal{F}}. \tag{12}$$

For recursive calculations put $K_k = cpT\binom{p-1}{k}Q_{p-1-k}$, $0 \le k < p$. Thus $I_q = (q-1)!K_{q-1}$. The factorial distinction is needed: the native Taylor coefficient contains $1/q!$, whereas conversion of a power channel contributes $(q-1)!$. Losing either factor changes the reconstruction formula.

6 Exact finite predictable innovations

Fix a grid and a polynomial P of total degree at most d. For cell j, put $Q_j = \Pi_{j+1}P$, and let R_j set $x_i = 0$ for $i \ge j$. Define the native coefficients

$$a_{j,q}(P) = \frac{1}{q!} R_j \partial_{x_j}^q Q_j. \tag{13}$$

They depend only on strict-past increments, vanish for $q > d$, and Lemma 3.1 gives

$$\|a_{j,q}(P)\|_{\mathcal{D}} \le \frac{e^T}{q!} \|P\|_{\mathcal{D}}. \tag{14}$$

Only the current and past increments survive in Q_j. Its exact Taylor expansion in the current increment therefore is $Q_j = \sum_{q=0}^{d} a_{j,q}(P) x_j^q$. Integrate that increment to obtain $\Pi_j P = \sum_{q=0}^{d} a_{j,q}(P)\mu_q(Th)$. Subtracting proves the cell identity

$$\Pi_{j+1}P - \Pi_j P = \sum_{q=1}^{d} a_{j,q}(P)\big(x_j^q - \mu_q(Th)\big). \tag{15}$$

The zeroth term vanishes because $\mu_0 = 1$. Summing over cells telescopes exactly:

$$P - \mathcal{E}_T P = \sum_{q=1}^{d} \sum_{j=0}^{N-1} a_{j,q}(P)\big(x_j^q - \mu_q(Th)\big). \tag{16}$$

No term has been discarded on a scaling heuristic. Every integral in this derivation is an actual finite signed polynomial integral.

Refine a fixed base polynomial P to successive grids and denote its refinement by P_n. Its position-coordinate norm is unchanged and its total degree stays at most d. Define the native field

$$A_{n,q}(P) = \sum_j W_j(a_{j,q}(P_n)). \tag{17}$$

Since cell lengths sum to one, (14) gives

$$\|A_{n,q}(P)\|_{\mathcal{F}} \le \frac{\|J_T\|\, e^T}{q!} \|P\|_{\mathcal{D}}. \tag{18}$$

This estimate is independent of degree and mesh, but does not by itself prove that the sequence converges. That requires the next two quantitative arguments.

7 Signed power estimates in the test dual

For strict-past H and integer $r \geq 0$, signed integration by parts gives the exact dual identity

$$J_T(Hx_j^{r+1}) = \sum_{k=0}^{p-1} (r)_k K_k(W_j(Hx_j^{r-k})). \tag{19}$$

Terms with $k > r$ vanish and can be omitted. To check the identity, test first against a polynomial G on a common finer grid. The coarse increment is the sum of fine increments in its cell. Apply (6) to each fine increment and use Leibniz's rule on $x_j^r G$. Derivatives of H vanish because it is strict past. The k derivatives on x_j^r give $(r)_k x_j^{r-k}$, the other derivatives give $D_s^{p-1-k}G$, and summing the fine durations gives exactly the cell integral in K_k. This proves (19) on every polynomial test. Density and continuity give the identity in $\mathcal{D}^*$.

Define nonnegative constants recursively by

$$B_0 = 1, \qquad B_{r+1} = pT \sum_{k=0}^{\min(r,p-1)} \binom{p-1}{k} (r)_k B_{r-k}. \tag{20}$$

They depend on p, T, r, not on the mesh or H.

Lemma 7.1. *For a strict-past polynomial H and a cell of normalized length $h \leq 1$,*

$$\left\| J_T(Hx_j^r) \right\| \leq B_r \left\| J_T(H) \right\|, \tag{21}$$

$$\left\| J_T(Hx_j^q) \right\| \leq B_q h \left\| J_T(H) \right\| \quad (q \geq 1), \tag{22}$$

$$\left\| J_T(Hx_j^q) \right\| \leq B_q h^2 \left\| J_T(H) \right\| \quad (q > p). \tag{23}$$

For $1 \leq q \leq p$,

$$\left\| J_T(Hx_j^q) - I_q(W_j H) \right\| \leq B_q h^2 \left\| J_T(H) \right\|. \tag{24}$$

Proof. The channel bound for a step is $\|K_k(W_j V)\| \leq pT\binom{p-1}{k} h \|J_T(V)\|$. Apply it in (19) and induct on $r+1$, using (20). This first proves (22); since $h \leq 1$, it also proves (21) and closes the induction. For $q > p$, every surviving index $k \leq p-1$ has $q-1-k \geq 1$. Use (22) on that residual power in each term of (19). The additional step length supplies h^2, proving (23). For $q \leq p$, the last term $k = q-1$ is exactly $(q-1)!K_{q-1}(W_j H) = I_q(W_j H)$. Every other term has a positive residual power, so the same argument bounds their sum by $B_q h^2 \|J_T(H)\|$. □

By (4), $\mu_q(Th) = 0$ for $1 \leq q < p$ and $\mu_p(Th) = cp!Th$. For $q > p$, $|\mu_q(Th)| \leq M_q h^2$ with a mesh-independent M_q: if $q = pk$, take $M_q = q!T^k/k!$ and note $k \geq 2$; otherwise take zero. Thus summing over $N = 1/h$ cells, under any uniform bound $\|J_T(H_j)\| \leq M$, gives the following three limits:

$$J_T\left(\sum_j H_j(x_j^q - \mu_q(Th)) \right) \longrightarrow I_q(A), \qquad 1 \leq q < p, \tag{25}$$

$$J_T\left(\sum_j H_j(x_j^p - \mu_p(Th)) \right) \longrightarrow 0, \tag{26}$$

$$J_T\left(\sum_j H_j(x_j^q - \mu_q(Th)) \right) \longrightarrow 0, \qquad q > p, \tag{27}$$

where the first line also assumes $\sum_j W_j H_j \to A$ in $\mathcal{F}$. The errors are $O(h)M$ in dual norm. For the critical line, (11) gives $I_p(W_j H_j) = cp!ThJ_T(H_j)$, which cancels the centering exactly. This is why the final centered reconstruction has $p-1$ channels.

8 Causal coefficients and convergence under refinement

It remains to prove the field convergence needed in (25). Introduce the causal comparison coefficient

$$b_{j,q}(P) = \frac{1}{q!}\Pi_j(\partial_{x_j}^q P), \tag{28}$$

and its step field $\widetilde{A}_{n,q} = \sum_j W_j(b_{j,q}(P_n))$. The derivative here is taken before the strict-past projection. It does not in general commute through that projection.

Lemma 8.1. *For every grid polynomial,*

$$b_{j,q}(P) = \frac{1}{q!}R_j e^{cTh\partial_{x_j}^p}\partial_{x_j}^q \Pi_{j+1} P. \tag{29}$$

Consequently, for a fixed cylinder under refinement,

$$\left\|\widetilde{A}_{n,q} - A_{n,q}\right\|_{\mathcal{F}} \le \frac{\|J_T\|\, e^T}{q!}(e^{Th}-1)\,\|P\|_{\mathcal{D}} \longrightarrow 0. \tag{30}$$

Proof. The current coordinate is retained by Π_{j+1}, so its derivative commutes with the future integrations. Integrating that one current coordinate after future conditioning is exactly heat evolution followed by clamping, by (5). This proves (29). The norm of $e^{cTh\partial_{x_j}^p} - \mathrm{id}$ is at most $e^{Th}-1$. Use contractivity of derivatives and clamping, the e^T projection bound, and the sum of normalized cell lengths equal to one. □

The heat correction proves equality of the eventual native and causal limits, but does not itself prove that either sequence has a limit. For that we need control of a moving conditioning cut.

Lemma 8.2. *For each degree bound d, there is $L_d \ge 0$, depending only on p, T, d, such that on every grid and every degree-at-most-d polynomial P,*

$$\|J_T(\Pi_b P - \Pi_a P)\| \le L_d(b-a)h\,\|P\|_{\mathcal{D}}, \qquad 0 \le a \le b \le N. \tag{31}$$

Proof. For each $q \ge 1$ there is $V_q \ge 0$ with $|\mu_q(Th)| \le V_q h$: use zero if $p \nmid q$, and $V_q = q!T^k/k!$ if $q = pk$, since $h^k \le h$. The cell identity (15), the positive power estimate, and (14) imply

$$\|J_T(\Pi_{j+1}P - \Pi_j P)\| \le h\,\|P\|_{\mathcal{D}}\,\|J_T\|\,e^T \sum_{q=1}^{d} \frac{B_q + V_q}{q!}.$$

Take the finite nonnegative sum as L_d, and sum the cell differences from a to $b-1$ by the triangle inequality. The argument proves dual-norm continuity of conditional increments; it does not assert continuity of moving-time positions in the strong factorial source norm. □

Proposition 8.3. *For each fixed cylinder P and every $q \ge 0$, the native fields $A_{n,q}(P)$ converge in $\mathcal{F}$ to a field $C_q P$. The causal fields converge to the same limit, and the bound* (1) *holds on the core.*

Proof. Compare adjacent grids. Each coarse increment splits into two fine increments. Differentiating the pulled-back polynomial in either fine coordinate is the pullback of differentiating in the coarse coordinate. Its degree stays at most $d = \deg P$, and its factorial norm after any number of such derivatives is at most $\|P\|_{\mathcal{D}}$.

At the first half of a coarse cell, the strict-past projections agree by signed convolution consistency. At the second half, the fine conditioning cut has advanced by just one fine cell. Apply (31) to the derivative polynomial. Dividing by $q!$ and integrating over the unit normalized horizon gives

$$\left\|\widetilde{A}_{n+1,q} - \widetilde{A}_{n,q}\right\|_{\mathcal{F}} \le \frac{L_d}{q!} 2^{-(n+1)} \|P\|_{\mathcal{D}}\,. \tag{32}$$

This bound is summable in n. Thus the causal fields are Cauchy in the complete space $\mathcal{F}$ and converge. Equation (30) gives the same limit for the native fields. The norm bound follows by passing to the limit in (18). Each native field is an adapted polynomial step field; closedness gives $C_q P \in \mathcal{A}_T$. □

For $q > d$ all native fields are zero, hence $C_q P = 0$. If two base-grid presentations define the same cylinder, their refinements agree on a common tail. Uniqueness of norm limits therefore makes the limit independent of the presentation. Finite-grid linearity and continuity of addition and scalar multiplication make $P \mapsto C_q P$ linear. These observations are necessary before extending the maps to completed observables.

9 The multichannel formula and observable completion

Proposition 9.1. *Every polynomial cylinder satisfies*

$$J_T(P - \mathcal{E}_T P) = \sum_{q=1}^{p-1} I_q(C_q P).$$

Proof. Refine the fixed polynomial to all sufficiently fine grids. Choose $d' \ge \max(\deg P, p-1)$ once and pad the exact telescope (16) to degree d' with zero coefficients. Refinement preserves the degree bound, so this is the same finite number of terms on every grid. The left side, after applying J_T, is the same dual element on each grid.

For each q, (14) gives a uniform bound on the paired native coefficients. Proposition 8.3 provides the L^1 convergence for the low powers. Apply (25) to $q < p$, the exact critical cancellation (26) to $q = p$, and (27) to $q > p$. Pass through the one fixed finite sum. The surviving terms are exactly those in the displayed identity. □

Proof of Theorem 1.1. The degree-independent bound (1) on the dense core extends each linear coefficient map uniquely to a bounded linear map $C_q : \mathcal{D} \to \mathcal{F}$ with the same bound. For a completed observable choose polynomials $P_k \to u$ in $\mathcal{D}$. Then $C_q P_k \to C_q u$. Closedness of $\mathcal{A}_T$ preserves adapted membership. The centered map $u \mapsto J_T(u - \mathcal{E}_T(u)1)$ is bounded linear, as is each $I_q C_q$. Thus Proposition 9.1 passes to the limit and proves (3) for every $u \in \mathcal{D}$.

For the simultaneous actual signed approximation, assemble the $p-1$ native fields and the native surviving power sum into one finite product Banach space. For each fixed P_k, the preceding arguments show convergence of that package as the mesh tends to zero. Its limiting package is the collection $(C_1 P_k, \dots, C_{p-1} P_k, J_T(P_k - \mathcal{E}_T P_k))$, which tends to the analogous package for u by continuity. Choose a mesh index n_k at least k and at least the base-grid index of P_k, large

enough that the package error is less than $1/(k+1)$. The triangle inequality gives simultaneous convergence to the package for u. Every approximating sum comes from the actual finite signed polynomial laws by (10). □

The diagonal is a proved selection; it is not every arbitrary joint degree/mesh limit. The convergence constants for one fixed polynomial may depend on its degree, as L_d shows. The extension to $\mathcal{D}$ instead uses the degree-independent coefficient bound. Those two kinds of bounds have distinct roles in the proof.

10 Uniqueness and the biharmonic case

The lower derivative orders cannot all be replaced by a coefficient depending only on the pth derivative. The terminal cylinder $P = Y_1$ has $D_s^p P = 0$ for every s, and $\mathcal{E}_T P = 0$. Nevertheless

$$J_T(P)(Y_1^{p-1}) = \mathcal{E}_T(Y_1^p) = cp!T \neq 0.$$

Thus any proposed reconstruction which vanishes whenever the pth derivative vanishes fails on this actual signed polynomial pairing. This elementary obstruction explains the role of lower channels without invoking a positive pseudo-L^2 interpretation.

If another bounded linear coefficient map agrees with C_q on all canonical polynomial limits, it equals C_q by density. This proves uniqueness of the coefficient extension. It does not assert that an arbitrary channel representation has unique integrands. Let

$$S : (\mathcal{F})^{p-1} \to \mathcal{D}^*, \qquad S(A_1, \dots, A_{p-1}) = \sum_{q=1}^{p-1} I_q A_q.$$

Then $S(A) = S(B)$ if and only if $A - B \in \ker S$. No theorem here sets that kernel equal to zero.

For $p = 4$ the three surviving channels are

$$\begin{aligned}
(I_1 A)(v) &= -4T \int_0^1 A(s)(D_s^3 v)\, ds, \\
(I_2 A)(v) &= -12T \int_0^1 A(s)(D_s^2 v)\, ds, \\
(I_3 A)(v) &= -24T \int_0^1 A(s)(D_s v)\, ds.
\end{aligned}$$

The native coefficient fields retain factors 1, 1/2!, and 1/3! respectively. The fourth channel is the drift $I_4 = -24TQ_0$ and cancels against the fourth moment centering. This example keeps the signs, factorials, and complementary test derivative orders visible.

For $p = 2$ the same construction has one surviving channel $I_1 = 2TQ_1$, with the kernel normalization of generator ∂_x^2. It is compatible with that Brownian variance convention, but the theorem's proof has not relied on positivity or on importing a probability-space martingale representation.

11 Verified source and literature boundary

The mathematical source is the frozen Git commit

`bab688bb6140cd6316c80680095a914ea0d346db.`

The namespaces in the source map share the following prefix:

`OperatorDerivativeRewrite.Foundation`

The file `theorem-map.json` records exact files, declarations, line numbers, source hashes and baseline GitHub links.

Content	*Lean namespace and principal declaration*
Domain and core	`HigherHeatAnalyticCylinderDomain.core_injective`, `core_dense`; `MultivariateFactorialDomain.derivative_norm`
Signed product pairing	`HigherHeatCompletedPairing.pairing_core_complex`, `pairing_not_positive`
Exact finite telescope	`HigherHeatMultitimeJetTelescope.telescope`, `native_dual_identity`
Power bounds	`HigherHeatAllPowerBounds.positive_increment_bound`, `higher_increment_bound`; `HigherHeatAllChannelRemainder.cell_remainder_bound`
Centered channel limits	`HigherHeatCenteredPowerLimits.low_centered_converge`, `critical_centered_vanish`, `higher_centered_vanish`
Native/causal correction	`HigherHeatCausalJetIdentification.physical_shifted_jet_eq`, `causal_field_approximation`
Cut and refinement bounds	`HigherHeatConditionalCutBound.conditional_cut_bound`; `HigherHeatCausalFieldCauchy.causal_fields_step_bound`
Field limits	`HigherHeatMultitimePredictableFields.field_tendsto`, `field_bound`
Polynomial reconstruction	`HigherHeatMultitimeReconstruction.polynomial_reconstruction`
Observable extension	`HigherHeatMultitimeFieldMaps.completed_bound`, `completed_mem`; `HigherHeatCompletedClarkOcone.clark_ocone`
Uniqueness	`HigherHeatCompletedClarkOcone.coefficient_unique`, `synthesis_uniqueness`
Actual joint diagonal	`HigherHeatSignedDiagonalRealization.completed_signed_realization`

The baseline receipt records a complete successful root build and a complete 54,943-declaration standard-axiom audit. Publication work changes no Lean source and reuses that verified baseline. The human proofs above reorganize the actual theorem mechanism; the notation B_r, L_d corresponds to recursive power bounds and finite degree-dependent constants in the formal sources. It does not claim a fresh Lean theorem merely because a constant has a shorter name in this exposition.

The primary comparison in [2], especially its discussion leading to Theorem 18, already retains lower power increments in a generalized Itô calculus for complex random-walk approximations. Its expected chain-rule formula is distinct from the present completed real signed-cylinder test-dual identity. That distinction is a comparison of the stated constructions, not a proof of priority for the present one. The preliminary literature check identifies earlier signed heat calculus and alternative complex-walk approaches. It has not established priority for this particular analytic completion or the multitime reconstruction theorem. Hochberg's paper was identified bibliographically; its complete text was not retrieved in this pass, so no uninspected theorem-number comparison is

made. The cited time-change paper is a primary contemporary reference for pseudoprocesses, not a premise extending the present result to signed random clocks. A publication decision requires a targeted comparison of domains and reconstruction topologies.

References

[1] K. J. Hochberg, A signed measure on path space related to Wiener measure, *Ann. Probab.* **6** (1978), 433–458.

[2] S. Bonaccorsi, C. Calcaterra and S. Mazzucchi, An Itô calculus for a class of limit processes arising from random walks on the complex plane, arXiv:1603.05437 (2016). `https://arxiv.org/abs/1603.05437`.

[3] F. Cinque and E. Orsingher, Higher-order fractional equations and related time-changed pseudo-processes, arXiv:2402.13691v2 (2024). `https://arxiv.org/abs/2402.13691`.